

Bridging The Gap: Semantic Vector Middleware For Knowledge Graph Construction

Abdul Haseeb Shams, Mohammad Bachcha, Ahmed Khaled
Department of Computer Science, Northeastern Illinois University

Doc2Graph

Core Objective:
Doc2Graph is a web based ingestion engine designed to transform semi-structured MongoDB documents into interactive and a connected Knowledge Graph. The challenge Doc2Graph aspires to solve is that NoSQL documents often contain deeply nested and inconsistent schemas that hide relational patterns. Doc2Graph automates the discovery of these patterns by mapping document hierarchies to graph nodes and edges

Scalable Clustering:
To manage large document collections, Doc2Graph implements a semantic grouping phase to avoid computational bottleneck of comparisons. Nodes are transformed into a high dimensional vector and the resultant embedding is processed through K-Means Clustering by grouping similar nodes, significantly narrowing the search space.

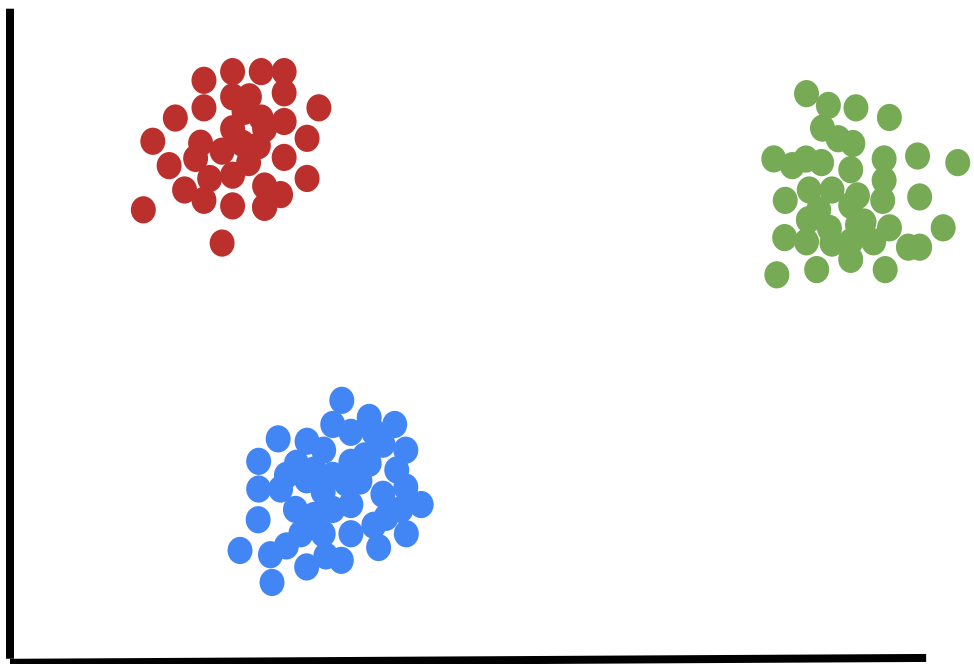


Chart 1: K means Clustering of Nodes

Semantic Merging:
A user Defined similarity threshold identifies and flags nodes within a cluster that are similar. The tool aggregates unique attributes from separate nodes into a single record, resolving data fragmentation before the data is exported for visual Knowledge graphs to view

Vector Middleware

Weaviate & JIT Schema:
Traditional Graphs require pre defined schema which fails with highly nested denormalized data. This pipeline dynamically generates schema on the basis of incoming metadata.

Recursive Cross Referencing:
To preserve the deep nested nature of denormalized data we implemented recursive cross referencing of nested Json objects in order to ensure the many to many relationship is preserved semantically

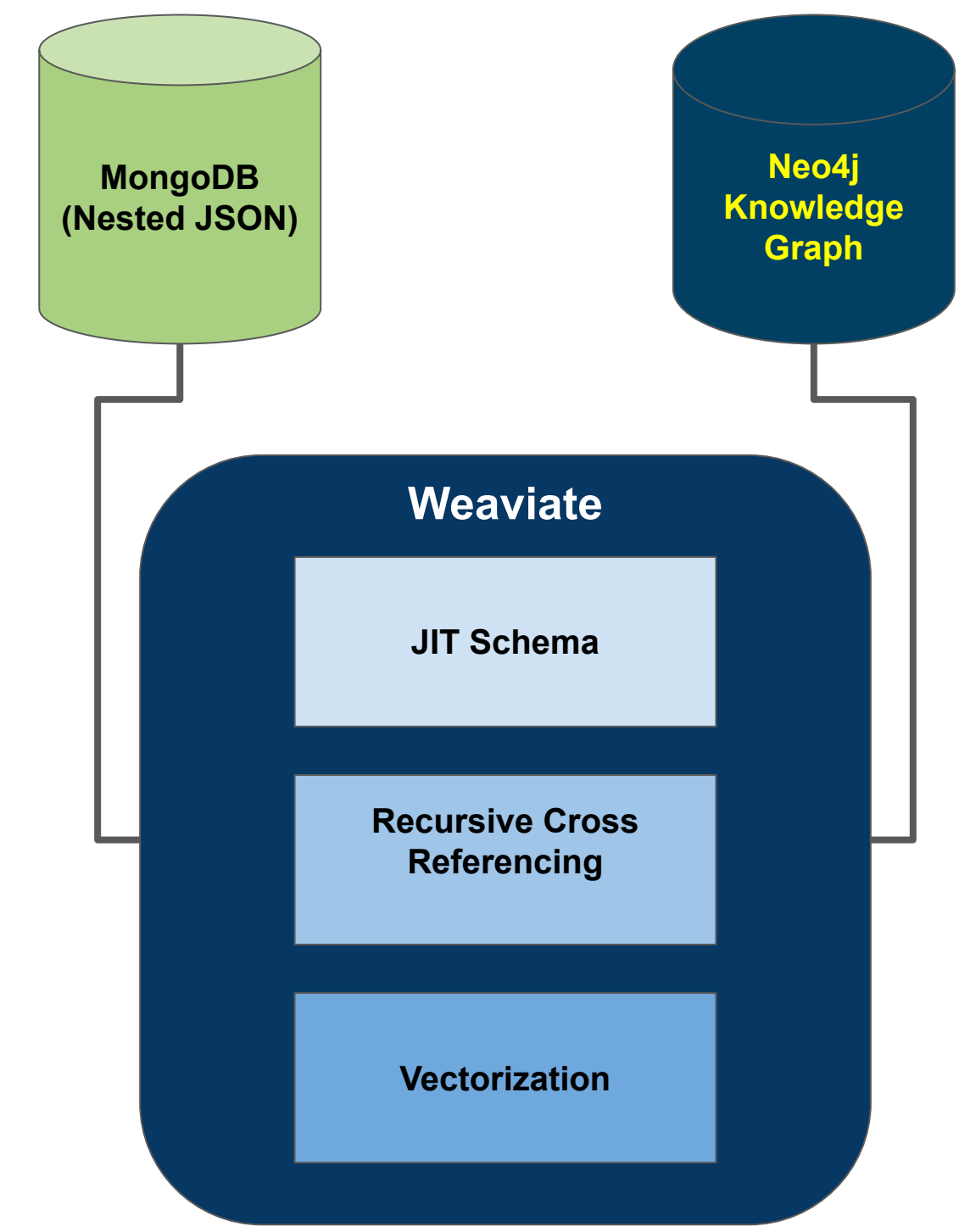


Figure 1: Automated ingestion pipeline utilizing weaviate for semantic resolution and dynamic relationship mapping

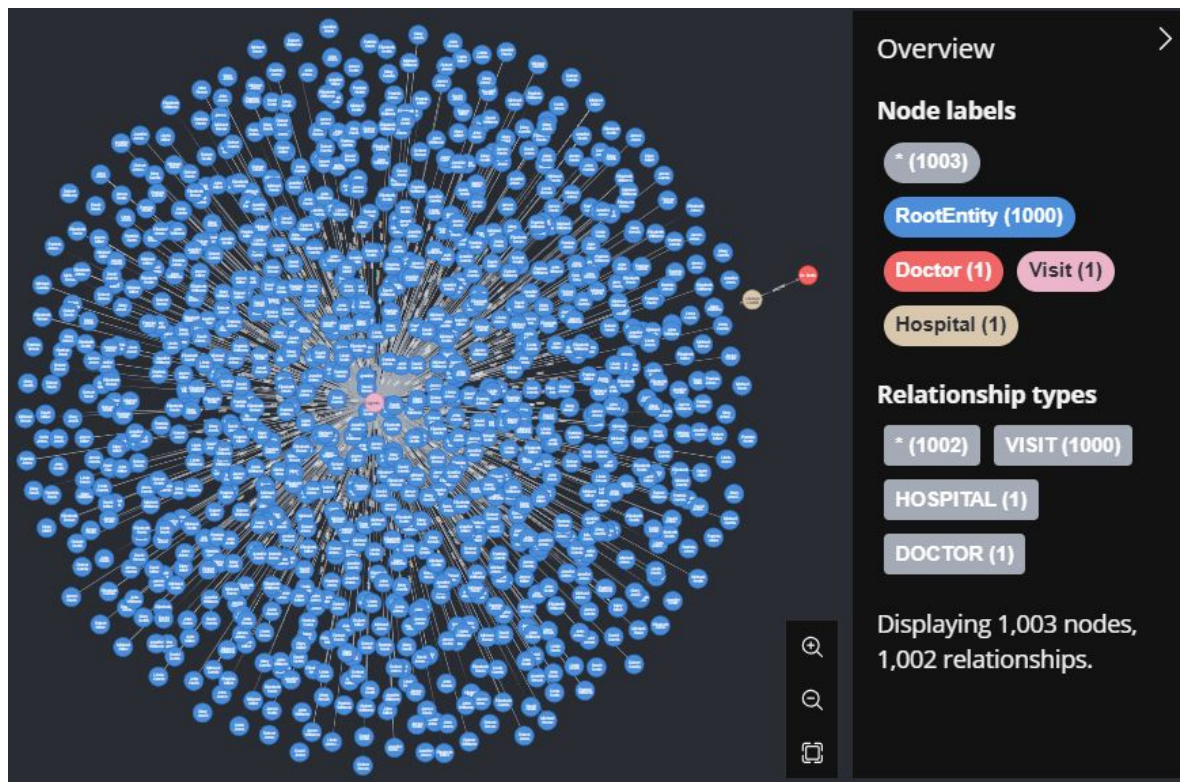


Figure 2: Visualizing Transition from Unstructured JSON to Normalized Knowledge Graph

Test Metric	
Weaviate (Nested)	
Ingestion latency (10,000 docs)	2407.97s
Deduplication %	99.95%
Neo4j Sync Success %	100
Neo4j sync throughput	29 records /s

Figure 3: Test Metric of Weaviate pipeline

Research Question

RQ1: Structural Integrity vs. Semantic Depth
Would flattening NoSQL data into a single vector provide better context at the cost of massive redundancy?

RQ2: Multi Stage Entity Resolution
How can we ensure the Perfect Duplicate detection without human intervention?

Future Work
Multi Engine benchmark:
The next phase will integrate Qdrant into middleware and evaluate comparative performance against flattened Json and nested Json

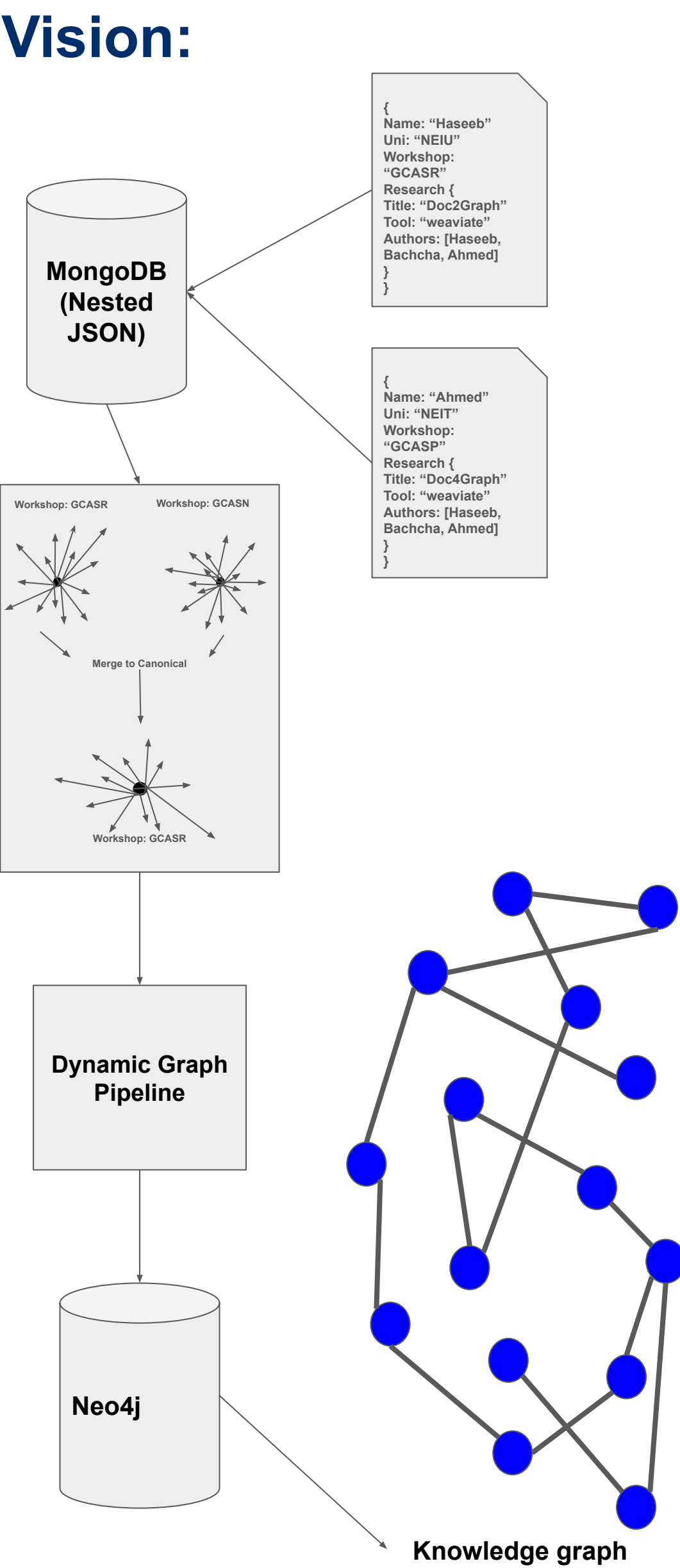


Figure 4: Vision for Semantic Vector Middleware for Knowledge Graph Construction